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Lab Manual

Experiment: File Handling in Python

Lab Title

File Handling in Python

Objective

After completing this lab, students will be able to:

- Understand the concept of file handling.
 - Create text files using Python.
 - Read data from files.
 - Write and append data to files.
 - Understand different file modes.
 - Use the `with` statement for safe file handling.
 - Apply file handling in real-world applications.
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Background

Programs often need to store information permanently. Variables can only store data while the program is running. Once the program ends, all variable data is lost.



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File handling allows Python programs to save information inside files so that it can be accessed later. This makes applications more practical because user information, reports, logs, and records remain available even after the program is closed.

Python provides built-in functions that make file handling simple and efficient.

Why Use File Handling?

File handling is used because it allows applications to store and retrieve information permanently.

Some common uses include:

- Saving user accounts
 - Login systems
 - Student record management
 - Employee databases
 - Sales reports
 - Attendance systems
 - Configuration files
 - Log files
 - Data analysis
-

Types of Files

Python mainly works with two types of files:

1. Text Files (.txt)

These files store readable text.



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Example:

Ali

Ahmed

Sara

Examples:

- info.txt
- students.txt
- notes.txt

2. Binary Files

These files store data in binary format.

Examples:

- Images
- Videos
- Audio
- PDF files

Opening a File

Python uses the `open()` function.

Syntax



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Python

```
file = open("filename.txt", "mode")
```

Example

Python

```
file = open("students.txt", "r")
```

File Modes

Mode	Description
r	Read only
w	Write (creates or overwrites file)
a	Append data



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x Create a new file

r+ Read and write

w+ Read and write (overwrite)

a+ Read and append

Closing a File

Always close files after use.

Syntax

```
Python  
file.close()
```

Example



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Python

```
file = open("students.txt", "r")  
  
print(file.read())  
  
file.close()
```

Using the with Statement (Recommended)

The `with` statement automatically closes the file after execution.

Syntax

Python

```
with open("students.txt", "r") as file:  
  
    print(file.read())
```

Advantages:

- Automatically closes files.
- Cleaner code.
- Prevents resource leaks.
- Recommended by Python developers.



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Writing to a File

Syntax

Python

```
with open("students.txt", "w") as file:  
    file.write("Ali")
```

Expected File Content

None

Ali

Writing Multiple Lines

Example



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Python

```
with open("students.txt", "w") as file:  
    file.write("Ali\n")  
    file.write("Ahmed\n")  
    file.write("Sara\n")
```

Expected File Content

None

Ali

Ahmed

Sara

Appending Data

Appending adds new data without removing previous content.

Example



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Python

```
with open("students.txt", "a") as file:  
    file.write("Ayesha\n")
```

Expected File Content

None

Ali

Ahmed

Sara

Ayesha

Reading Entire File

Example

Python

```
with open("students.txt", "r") as file:
```



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```
data = file.read()  
print(data)
```

Expected Output

```
None  
Ali  
Ahmed  
Sara  
Ayesha
```

Reading Line by Line

Example



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Python

```
with open("students.txt", "r") as file:  
    for line in file:  
        print(line.strip())
```

Expected Output

None

Ali

Ahmed

Sara

Ayesha

Reading Using readlines()

Example



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Python

```
with open("students.txt", "r") as file:  
    lines = file.readlines()  
print(lines)
```

Expected Output

None

```
['Ali\n', 'Ahmed\n', 'Sara\n', 'Ayesha\n']
```

Practical Example 1

Saving Student Information

Python

```
name = input("Enter Name: ")  
roll = input("Enter Roll Number: ")
```



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```
with open("students.txt", "a") as file:  
    file.write(name + "," + roll + "\n")  
print("Record Saved Successfully.")
```

Sample Input

None

Ali

101

Expected File Content

None

Ali,101



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Practical Example 2

Reading Student Records

Python

```
with open("students.txt", "r") as file:  
    print(file.read())
```

Expected Output

None

Ali,101

Ahmed,102

Sara,103

Common File Handling Errors

FileNotFoundError



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Occurs when the specified file does not exist.

Example

Python

```
with open("abc.txt", "r") as file:  
    print(file.read())
```

PermissionError

Occurs when the program does not have permission to access the file.

IsADirectoryError

Occurs when a folder is opened as a file.

Best Practices

- Always use the `with` statement.
- Close files properly if not using `with`.
- Use meaningful file names.
- Handle exceptions where necessary.
- Never overwrite important files accidentally.



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-
- Store records in a structured format.
-

Lab Task 1

Login / Signup System

Objective

Create a Signup program that stores user information inside **info.txt**.

Requirements

Ask the user to enter:

- Full Name
- Username
- Email
- Password

Store the information in **info.txt** using comma-separated values.

Example File Content

None

Ali Khan,alikhan,ali@gmail.com,12345

Sara Ahmed,sara,sara@gmail.com,abc123



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Lab Task 2

Student Marks Record System

Objective

Create a program that stores student records.

Requirements

Take input:

- Student Name
- Roll Number
- Programming Marks
- Mathematics Marks
- AI Marks

Store the data inside **marks.txt** in comma-separated format.

Example

None

Ali, 101, 87, 92, 89

Sara, 102, 91, 90, 95

Ahmed, 103, 75, 81, 79



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Lab Task 3

Daily Notes Manager

Objective

Create a simple notes application.

Requirements

- Ask the user to enter today's note.
- Save the note in **notes.txt**.
- Every new note should be appended.
- After saving, display all notes stored in the file.

Example Output

Today's Notes

None

Complete Python Assignment

Study File Handling

Prepare Lab Manual



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Conclusion

In this lab, students learned how to create, read, write, and append text files using Python. They explored different file modes, understood the advantages of using the `with` statement, practiced reading files in multiple ways, and implemented real-world applications such as a signup system, a student marks recorder, and a notes manager. These concepts form the foundation for building applications that require permanent data storage.